

Response to the Referee

Principles for Open Data Curation: A Case Study with the New York City 311 Service Request Data

We thank the Editor and the Referee for their careful reading and helpful comments, and we appreciate the Referee’s positive assessment of the paper. We have revised the manuscript in response; the main changes are:

1. **Figures.** Figures have been revised, replaced, or removed to improve readability, with larger fonts, explicit units, and expanded captions that state each figure’s key observations. One figure has been replaced by a table, and figures and tables have been renumbered accordingly.
2. **Durations.** We added an explanation of why many long durations are plausible given the agency and complaint type, and we now summarize negative durations numerically in the text.
3. **Data curation roles.** Following the Referee’s suggestion, the Collaboration step now recommends formalizing a data curation responsibility within each agency.
4. **Clarity and corrections.** We added examples of redundant fields, removed a paragraph on embedded commas after verifying that the export handles them correctly, and corrected the typographical issues the Referee identified.

Point-by-point responses follow, with the Referee’s comments in italics. Page and line numbers refer to the revised manuscript. Where a figure has been renumbered, we give both its original and revised numbers. Tables in this letter are numbered R1 and R2 to distinguish them from tables in the manuscript.

Responses to the Referee

Comment 1.

Page 4, lines 31–32: The authors mention two potentially redundant pairs. This is discussed later, but it might help to add something like “e.g., cross_street and intersection_street” at this point in the article.

Response.

We thank the Referee for this suggestion, which improves the clarity of the paragraph. We now give the `cross_street` and `intersection_street` pairs as examples at first mention (p. 4, ll. 8–9), which now reads: “The dataset contains seven street-related fields, including two likely redundant pairs, e.g., `cross_street_1` & `intersection_street_1` and `cross_street_2` & `intersection_street_2`.”

Comment 2.

Page 4, bottom: This is just a comment for what it’s worth. I was doing text analysis on data like this that contained a Notes field. It was a CSV file, and I imported it into MATLAB. I ended up with a strange number of columns when importing it. It turns out that the commas inside the Notes field were telling MATLAB that there were multiple columns from the Notes. So, CSV can be dangerous or problematic with free-form text

fields. You are alluding to problems like this at the bottom of the page. Just wanted to pass along my experience.

Response.

We thank the Referee for sharing this experience, which prompted us to explore and verify how the NYC Open Data portal handles embedded commas. The portal runs on the Socrata platform (now part of Tyler Technologies), a mature and widely used infrastructure. Socrata’s documentation states that “strings containing embedded commas, newlines, or quotes will themselves be quoted,” with embedded quotes escaped by doubling (Tyler Technologies, 2026).

This behavior matches RFC 4180 (Shafranovich, 2005), Rule 6: “Fields containing line breaks (CRLF), double quotes, and commas should be enclosed in double-quotes.” Although RFC 4180 is an informational memo rather than a formal standard, it documents common practice and is the closest thing to an official specification for the CSV format; it is widely used as the reference for CSV parsers and validators. It was published by the Internet Engineering Task Force (IETF), the body responsible for core internet protocols and formats.

Our examination of the raw CSV file exported from the NYC Open Data portal confirms this behavior. For example, the `location` field contains a comma by design, separating latitude and longitude, and appears in the raw file enclosed in double quotation marks:

"(40.64224331435751, -74.00707333743664)"

In order to explore the Referee’s concern further, we scanned the full, unmodified export (approximately 16.0 million records; NYC Office of Technology and Innovation, 2026) to check its compliance with RFC 4180. We found embedded commas in 15 of the 41 fields; Table R1 summarizes the results.

Table R1: Fifteen fields in the raw 311 SR CSV export (≈ 16.0 million records) contain embedded commas. There are none in the remaining 26 of 41 fields. No field contains an embedded line break. All values are properly quoted, and every record parses to 41 columns.

Field	Values with commas	% of records	Values with quotes	Example
<code>location</code>	15,713,332	98.11	–	"(40.6409..., -73.9736...)"
<code>resolution_description</code>	2,371,902	14.81	160,327	<i>free-text narrative</i>
<code>descriptor</code>	138,943	0.87	2,099	"NOISE, BARKING DOG (NR5)"
<code>taxi_pick_up_location</code>	136,328	0.85	–	"2 AVENUE AND EAST 57 STREET, MANHATTAN, ..."
<code>park_facility_name</code>	1,628	0.01	233	"CAPTAIN JOHN MCKENNA, IV PARK"
<code>cross_street_2</code>	284	<0.01	–	"HUDSON LINES,"
<code>road_ramp</code>	222	<0.01	–	"WTC / PATH (SOUTH END, FROM ..."
<code>bridge_highway_name</code>	135	<0.01	–	"DR. MARTIN LUTHER KING, JR. EXPWY"
<code>bridge_highway_segment</code>	82	<0.01	–	"JCT W/CROSS BRONX EXPWY EXTENSION (I-295), ..."
<code>incident_address</code>	22	<0.01	–	"534 NEW PALTZ RD, HIGHLAND, NY 12528"
<code>street_name</code>	21	<0.01	–	"NEW PALTZ RD, HIGHLAND, NY 12528"
<code>location_type</code>	3	<0.01	–	"1-, 2- AND 3- FAMILY HOME"
<code>city</code>	3	<0.01	–	"ASBURY PARK,"
<code>cross_street_1</code>	1	<0.01	–	"(S1)SHERIDAN EXPY, TRIBORO B"
<code>bridge_highway_direction</code>	1	<0.01	–	"S TO FRANKLIN AV, PROSPECT PARK"
Total (15 fields)	18,362,907		162,659	

Besides `location`, which always contains a comma, over 2.6 million values contain commas, mostly

in the free-text `resolution_description` field ($\approx 15\%$ of records), and $\approx 163,000$ values contain escaped double quotes. Nevertheless, every record parses to the expected 41 columns with the `fread()` function from the `data.table` package, confirming that all embedded commas and quotes are correctly quoted and escaped.

The Referee’s experience is therefore highly relevant: a parser that does not honor CSV quoting would misread a substantial share of these records. Because the Socrata export quotes these fields correctly, and quote-aware readers such as `fread()` (Barrett et al., 2025) interpret that quoting correctly, all fields are read as intended.

We have therefore removed the paragraph on this issue (formerly p. 4, ll. 45–46), as the problem it anticipated does not arise with these data.

Comment 3.

Figure 2: Please add more explanation to the caption. What do you want the reader to notice or take away from the figure? It’s ok to repeat what you have in the body of the paper.

Response.

We thank the Referee for this suggestion. In revising the caption, we concluded that the figure contained more detail than a reader could readily absorb, and that its key points are conveyed more clearly in the text. We have therefore removed the figure and revised the accompanying discussion (p. 4, ll. 27–30) to state its main takeaway directly: just six agencies account for 90.19% of all SRs.

Comment 4.

Figure 3: First, this is very hard to read. The fonts are too small. You could do the same as Figure 2 and just show the top 90%. I also make the same suggestion as Figure 2, that you add more details and explanations to the caption, so the figure stands alone from the text in terms of what you want readers to know.

Response.

We agree, and we thank the Referee for the suggestion to show only the top contributors. We have replaced the figure with a table (Table 1 of the revised manuscript, p. 4) listing the six agencies that together account for approximately 90% of all SRs. The table is set in the body font, which resolves the readability issue, and its caption states the key observation: SR volume is highly concentrated, with just six agencies generating the large majority of 311 SRs.

Comment 5.

Figure 4: Are the units for both of these plots days? You should specify the units. I guess the violin plots show Number of SRs, which should be made clearer. I think the one on the left mentions days as ‘d’, but I would spell it out. Also, these might be more readable with the plot placed vertically.

Response.

We thank the Referee for these observations. The duration axis is in days, and the violin plot shows the number of SRs; both axes are now labeled explicitly, and “d” has been spelled out as “days” throughout. To improve readability, we removed panel (b), which showed negative durations; those

cases are now summarized numerically in the text (p. 12, ll. 13–23). The remaining plot now spans the full page width. The revised caption explains how to read the plot and states its key observations. (Figure 4(a) in the original manuscript is Figure 1 in the revision.)

Comment 6.

Page 11, line 26: You have 30 d. I would spell it out and use 30 days.

Response.

Corrected. The text now reads “more than 30 days post-closure” (p. 9, ll. 33–35). The abbreviation arose from the `siunitx` unit `\day`, which prints as “d”; we have replaced it with plain text throughout the document.

Comment 7.

Page 11, line 34: You have ‘datetime’. Suggest using date-time or date/time.

Response.

We agree. On review, we removed the sentence containing this instance (p. 11, l. 34 of the original manuscript), as its description of SQL date handling was imprecise. The remaining instances of “datetime” have been changed to “date-time” (p. 10, ll. 37–38).

Comment 8.

Figure 5: These are impossible to read, because the font is so small. Suggest you place these vertically and make the fonts larger. Also, indicate what the red dashed line represents.

Response.

We agree. The two panels are now stacked vertically at full page width, with enlarged axis labels, tick labels, and titles. The red dashed line marks three standard deviations above the mean hourly SR count ($\mu + 3\sigma$); it is now labeled directly on the plot, drawn more prominently, and defined in the caption, which also summarizes the figure’s key observations. (Figure 5 in the original manuscript is Figure 2 in the revision.)

Comment 9.

Page 12, line 21: You need a space between the and “closed...”

Response.

Corrected (p. 10, l. 34).

Comment 10.

Page 13, positive durations: Do the long durations make sense given the agency? If so, some words here might help readers.

Response.

We thank the Referee for this question, which required some additional analysis. We extracted the SRs with long (and extreme) durations and examined their associated agencies and `complaint_type`

values. Table R2, at the end of this letter, summarizes the results by agency, showing the average duration and count for each complaint type.

Most of these long-duration SRs are associated with complaints that could plausibly take a long time to resolve. For example, many of the Department of Parks and Recreation (DPR) SRs concern trees (new tree requests and overgrown, damaged, or dead trees), which can require lengthy efforts to plant a new tree or remove a dead or dying one. Likewise, the Department of Buildings (DOB) handles SRs involving plumbing, building use, and construction, which may also take considerable time to resolve (p. 13, l. 4–8).

Accordingly, we have revised the manuscript to include similar examples and to note that such SRs can reasonably require long resolution times.

Comment 11.

Page 17, Actionable Steps: I really liked these. I have one suggestion (take it or leave it) on Collaboration. Agencies could (should?) appoint a Data Curation Officer.

Response.

We thank the Referee for the kind words and for this suggestion, which we have adopted. NYC’s Open Data Law, Local Law 11 of 2012 (New York City Council, 2012), already requires each agency to designate an Open Data Coordinator. However, the Coordinator’s formal responsibilities, as set out in the City’s Open Data Technical Standards Manual (City of New York, 2026), focus on publishing, compliance reporting, and public outreach; none addresses data quality or curation. In our experience, the role is also typically assigned in addition to staff members’ core duties, with limited training and frequent turnover.

We have expanded the Collaboration step (p. 16, l. 46 to p. 17, l. 12) to describe this existing structure and to recommend formalizing a data curation responsibility, either by extending the Coordinator role or by appointing a dedicated Data Curation Officer, as the Referee suggests.

Table R2: Sample Complaint Types with long average positive durations, by agency. Durations are the mean number of days from creation to closure.

Agency	Complaint type	Avg. duration (days)	Count
DEP	SEWER	1,093	245
	WATER SYSTEM	947	71
DOB	REAL TIME ENFORCEMENT	987	50
	SPECIAL PROJECTS INSPECTION TEAM (SPIT)	975	91
	GENERAL CONSTRUCTION/PLUMBING	930	1,173
	PLUMBING	924	97
	BUILDING/USE	916	767
DOE	SCHOOL MAINTENANCE	931	189
DOHMH	POISON IVY	1,358	181
	WINDOW GUARD	1,269	78
	DAY CARE	1,220	4,087
	MOBILE FOOD VENDOR	1,206	6,489
	TATTOOING	1,172	505
	FOOD ESTABLISHMENT	1,162	20,906
	SMOKING	1,129	8,249
	NON-RESIDENTIAL HEAT	1,071	2,063
	DRINKING WATER	797	92
	INDOOR SEWAGE	796	113
DOT	STREET LIGHT CONDITION	1,064	1,204
	TRAFFIC SIGNAL CONDITION	866	471
DPR	NEW TREE REQUEST	1,162	28,262
	OVERGROWN TREE/BRANCHES	1,064	12,412
	DAMAGED TREE	1,017	4,491
	WOOD PILE REMAINING	997	198
	ROOT/SEWER/SIDEWALK CONDITION	994	6,899
	ILLEGAL TREE DAMAGE	972	674
	DEAD/DYING TREE	968	2,167
	UPROOTED STUMP	955	151
	MAINTENANCE OR FACILITY	929	1,122
DSNY	GRAFFITI	1,135	2,503
	ADOPT-A-BASKET	1,122	301
EDC	NOISE – HELICOPTER	1,041	32,529
HPD	HEAT/HOT WATER	1,088	64
	PAINT/PLASTER	963	52
	UNSANITARY CONDITION	959	56
	PLUMBING	952	52
	GENERAL	898	53
TLC	FOR HIRE VEHICLE COMPLAINT	942	75

References

- Barrett, T., M. Dowle, A. Srinivasan, et al. (2025). *data.table: Extension of ‘data.frame’*. R package.
- City of New York (2026). NYC Open Data Technical Standards Manual: City policies. <https://cityofnewyork.github.io/pendatatsm/citypolicies.html>. Accessed September 2026.
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- NYC Office of Technology and Innovation (2026). 311 service requests from 2010 to present. NYC Open Data, dataset erm2-nwe9, <https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9>. Accessed September 2026.
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